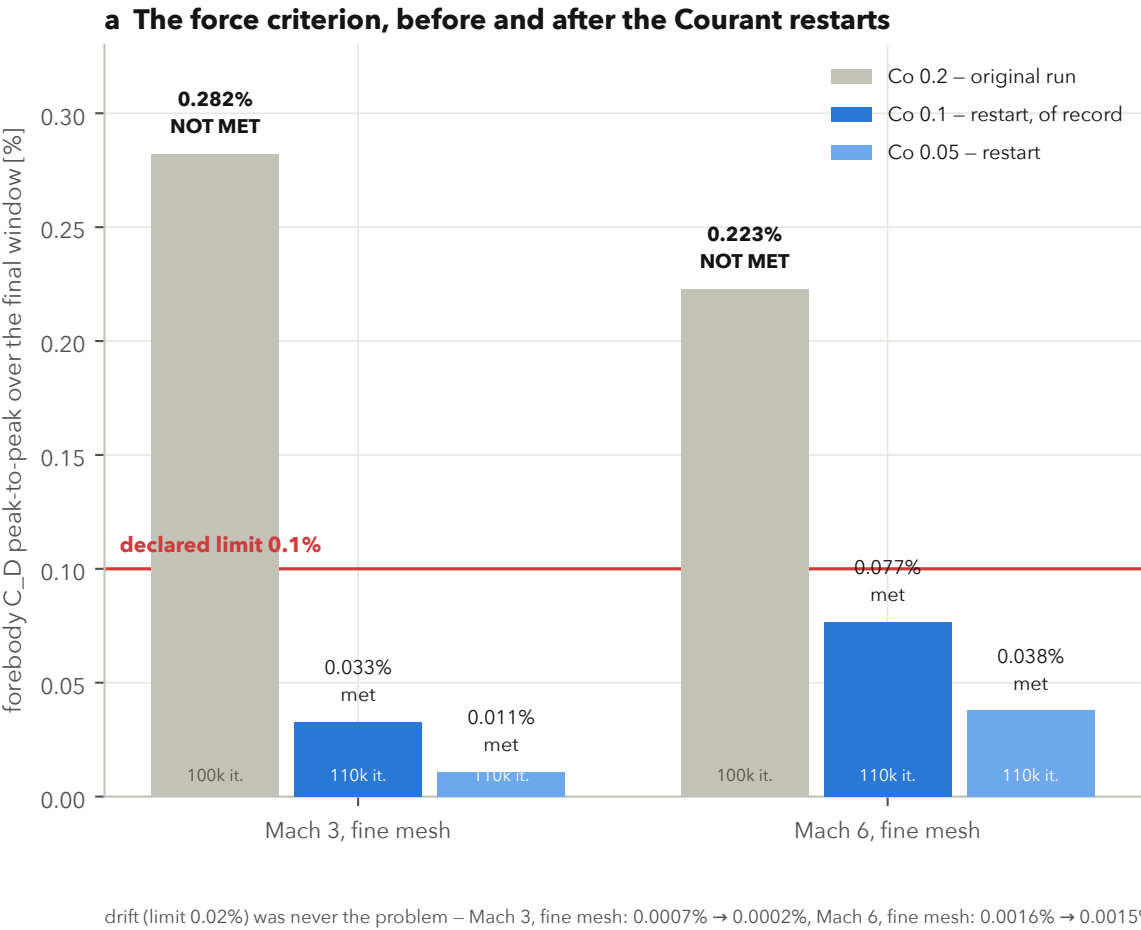
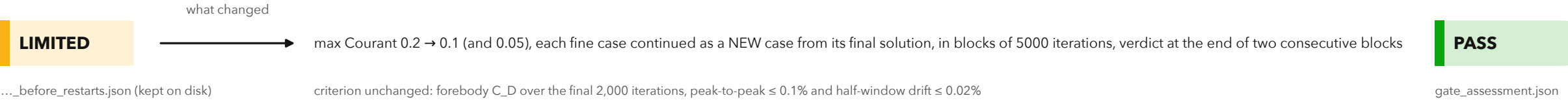


# 7 Gate G4: LIMITED → PASS, with the rule that got it there disclosed

Both fine-mesh sphere cases ended in a bounded limit cycle and missed the force criterion. The criterion was not moved; the Courant number was. The restart rule was written after the first results had been seen.



## b What the restart moved, and what it did not

|                               | Mach 3                      | Mach 6                      |
|-------------------------------|-----------------------------|-----------------------------|
| fine-mesh C_D, of record      | 0.84721 → 0.84692 (-0.035%) | 0.87221 → 0.87199 (-0.026%) |
| observed order of convergence | 0.67 → 1.13                 | 0.60 → 0.88                 |
| GCI, fine mesh                | 0.27% → 0.10%               | 0.36% → 0.19%               |
| fine – medium C_D difference  | 0.128% → 0.093%             | 0.151% → 0.125%             |

The mean of the limit cycle was not the fixed point: the converged C\_D sits about 0.03% off the oscillating window mean – small against every tolerance, but a third of the fine-medium difference, which is why the observed order and the GCI moved.

### DISCLOSURE (NR-25)

The restart rule, the two-block minimum and the rule for which solution is “of record” (the largest Courant number that met the criterion) were all written AFTER the original results had been seen; the two-block minimum was added while block 1 was running. Each makes the test stricter or is outcome-neutral, but none is pre-declared.

A reader who does not accept a rule written after the fact should read G4 as LIMITED; the evidence for both readings is in the run directory. The Mach 6 margin is thin, and went up between blocks.